A Non-Ergodic Lattice Framework for Joint Routing and Load Optimization

Abstract

We present a formal mathematical framework for the Damm Smart Truck challenge. By mapping the cargo space to a discrete lattice and the loading/unloading operations to sequential growth/ablation processes, we unify spatial and temporal routing constraints. We derive the conditions under which permutation operators preserve causal topology and demonstrate that the phase space of valid configurations is non-ergodic.

1 System Space

Let the truck's cargo volume be represented by a bounded spatial lattice subgraph $G = (V, E)$ where $V \subset \mathbb{Z}^3$. Let $n = |V|$ be the total number of loadable units.

The logistical operation is parameterized by a bijective sequence $\sigma : \{1, \ldots, n\} \to V$, representing the temporal order of loading.

2 Topological Causality and the Dependency Poset

Damm's trucks utilize lateral tarps, meaning pallets are accessed from the y-boundaries of the lattice.

Let $\mathcal{O}_\sigma(t)$ be the set of unoccupied lattice coordinates at time t during the unloading process:

$$\mathcal{O}_\sigma(t) = \mathcal{O}_0 \cup \bigcup_{k=1}^{t} \{\sigma(k)\} \tag{1}$$

where $\mathcal{O}_0$ contains the points in $\mathbb{Z}^3$ which are adjacent to the sides.

Definition 1 (Lateral Accessibility)**.** *A vertex $v \in V$ is accessible for loading at time t if it is geometrically adjacent to a previously loaded vertex or lies on the base/lateral boundary ∂V:*

$$acc_\sigma(v, t) \iff v \in \partial V \vee \exists (v, v') \in E \mid v' \in \mathcal{O}_\sigma(t-1) \tag{2}$$

A sequence σ is physically valid if and only if:

$$\forall t \in \{1, \ldots, n\}, \quad \mathrm{acc}_\sigma(\sigma(t), t) \text{ is True.} \tag{3}$$

This adjacency rule induces a strict partial order (a poset) over the vertices, denoted as $(V, \prec_c)$. We write $v \prec_c u$ if v must be loaded before u to provide the necessary support or topological link to the boundary.

3 Derivation of the Commutation Constraints

To optimize warehouse packing efficiency, an algorithm must explore the sequence space by applying a permutation operator $\hat{P}_{A,B}$ which swaps the loading indices t_A and t_B (where $t_A < t_B$). Let $\sigma' = \hat{P}_{A,B}\sigma$.

We derive the conditions under which $\hat{P}_{A,B}$ commutes with the topological validity operator $\hat{V}$, such that if $\hat{V}\sigma = \text{Valid}$, then $\hat{V}(\hat{P}_{A,B}\sigma) = \text{Valid}$.

Lemma 1 (Topological Independence)**.** *For the transposition to yield a valid sequence, the intermediate sequence must not depend causally on $\sigma(t_A)$.*

Proof. Let $v_A = \sigma(t_A)$ and $v_B = \sigma(t_B)$. Upon transposition, v_A is absent from the occupied set for all times $t \in (t_A, t_B)$. Therefore, the accessibility of any intermediate vertex $u = \sigma(t)$ must not rely solely on the edge (u, v_A).

$$\forall t \in (t_A, t_B), \quad \exists (v, \sigma(t)) \in E \mid v \in \mathcal{O}_\sigma(t) \setminus \{v_A\} \tag{4}$$

If this holds, the causal poset is preserved: $v_A \not\prec_c \sigma(t)$ for all intermediate t, and the commutation $[\hat{P}_{A,B}, \hat{V}] = 0$ is satisfied topologically. □

Simultaneously, we define an evaluation function $f(v) \in \mathcal{C}$, mapping a vertex to a Client ID. The delivery sequence defines temporal epochs $\{T_1, T_2, \ldots, T_c\}$.

Lemma 2 (Algebraic Invariance)**.** *For the external route distance to remain invariant under the swap, the transposition must preserve the sequence of client extractions.*

Proof. The transposition $\hat{P}_{A,B}$ preserves the client extraction sequence if and only if:

$$f(\sigma(t_A)) = f(\sigma(t_B)) \quad \vee \quad \exists k \mid T_k \leq t_A < t_B < T_{k+1} \tag{5}$$

This indicates that either the swapped products belong to the identical client (intra-epoch), or they are identical products destined for different clients, maintaining the macroscopic f-counts. □